

Surname / Van:

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FSK 176

Semester Test 2 - Semestertoets 2

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date / Datum: 2/11/2016

Time / Tyd: 2 hours / ure

Total / Totaal: 60 max

Examiner / Eksaminator: Dr J M Nel

Moderator / Moderator: R Q Odendaal

Test register: Tear carefully on the line above and hand to the invigilator.
Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 176: Semester Test 2 - Semestertoets 2

2/11/2016

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Please read the following carefully. Lees die volgende noukeurig

1. Answer all the questions in the space provided on the question paper. Antwoord alle vrae in die spasie gelaat op die vraagstel.

2. No pencil work will be marked. Do not use "tippex". Geen potlood-werk sal gemerk word nie. Moet nie uitwis-velcistof ("tippex") gebruik nie.

3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.

4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). Teken diagramme waar nodig (bv. vryliggaams-diagramme).

5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks. U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word!

6. Units must be used when numbers are substituted else marks will be subtracted. Een-hede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.

7. The use of the cheat sheet is only allowed subject to the rules stated on it. Die formuleblad mag slegs gebruik word onderheving aan die reëls daarop gedruk.

8. All test/exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. Alle toets/eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.

Constants and formulas / Konstantes en formules

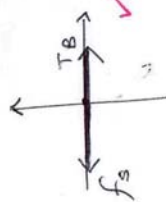
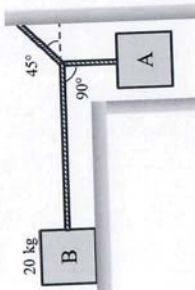
$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J/(mol}\cdot\text{K)}$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$; $r_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$; $c_{\text{ice}} = 2090 \text{ J/kg}\cdot^\circ\text{C}$;
 $c_{\text{water}} = 4186 \text{ J/kg}\cdot^\circ\text{C}$; $c_{\text{steam}} = 2010 \text{ J/kg}\cdot^\circ\text{C}$; $L_{\text{water, f}} = 3.33 \times 10^5 \text{ J/kg}$; $L_{\text{water, v}} = 2.26 \times 10^6 \text{ J/kg}$;
 $\rho_{\text{water, 0}^\circ\text{C}} = 0.999 \text{ g/cm}^3$; $\rho_{\text{water, 4}^\circ\text{C}} = 1.000 \text{ g/cm}^3$

1 Question 1 / Vraag 1

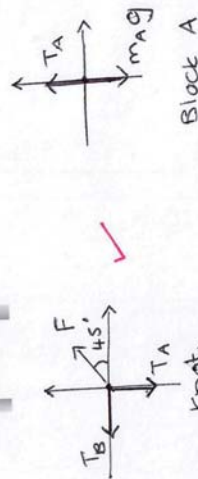
[10]

Block B rests on a surface for which the static and kinetic coefficients of friction are 0.60 and 0.40, respectively. The ropes are massless. What is the maximum mass of block A for which the system is in equilibrium?

Blok B rus op 'n oppervlak met statiese- en kinetiese-rywingskoeffisiënte van 0.60 en 0.40, onderskeidelik. Die toue is masseloes. Wat is die maksimum massa van blok A sodat die stelsel nog in ewewig sal wees?



Block B



Block A

* Block B: $\sum F_{Bx} = 0 = T_B - \mu_s N_B$ (fs) — ①
 $\sum F_{By} = 0 = N_B - m_B g$ — ②
 $\therefore N_B = m_B g$

From ① and ② $T_B = \mu_s N_B = \mu_s m_B g$

* Block A: $\sum F_{Ay} = 0 = T_A - m_A g$
 $\therefore T_A = m_A g$

* The knot: $\sum F_{x, \text{knot}} = 0 = F \cos 45^\circ - T_B$
 $\therefore F = \frac{T_B}{\cos 45^\circ} = \frac{\mu_s m_B g}{\cos 45^\circ}$

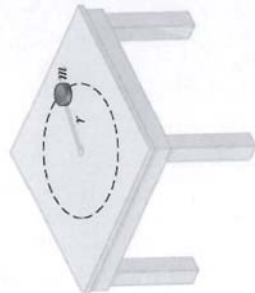
$\sum F_{\text{knot}, y} = 0 = F \sin 45^\circ - T_A$
 $= \left(\frac{\mu_s m_B g}{\cos 45^\circ} \right) \sin 45^\circ - m_A g$
 $m_A = \left(\frac{\mu_s m_B}{\cos 45^\circ} \right) \sin 45^\circ = \mu_s m_B \tan 45^\circ$
 $m_A = 0.6 (20 \text{ kg}) (1)$
 $= 12 \text{ kg}$

2 Question 2 / Vraag 2

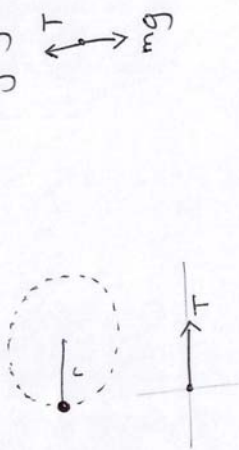
[5]

A light string can support a stationary hanging load of 25 kg before breaking. An object of mass $m = 3.00 \text{ kg}$ attached to the string rotates on a frictionless table in a circle of radius $r = 0.800 \text{ m}$ and the other end of the string is held fixed on the tabletop, as shown in the figure. What is the maximum speed that the object can have before the string breaks?

'n Ligte tou kan 'n stilstandende hangende las van 25 kg dra voordat dit breek. 'n Voorwerp met 'n massa $m = 3.00 \text{ kg}$ word aan die tou vasgemaak, en dit roteer in 'n sirkel met radius $r = 0.800 \text{ m}$ op 'n wrywinglose tafel. Die ander punt van die tou is vas aan die tafel, soos aangedui in die figuur. Wat is die maksimum speed wat die voorwerp kan hê voordat die tou breek?



Hanging mass



For the hanging mass: $\sum F_y = 0 = T - m_{\text{max}} g$
 $\therefore T = m_{\text{max}} g$
 $= 25 \text{ kg} (9.8 \text{ m/s}^2)$
 $T_{\text{max}} = 245 \text{ N}$

For circular motion on table:

$$\Sigma F = ma_c = m \frac{v^2}{r} = T \quad \checkmark$$

$$v_{\max}^2 = \frac{r T_{\max}}{m} \quad \checkmark$$

$$v_{\max} = \sqrt{\frac{r T_{\max}}{m}}$$

$$= \sqrt{\frac{(0.80m)(245N)}{3.00 \text{ kg}}}$$

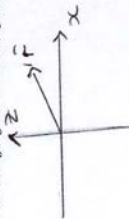
$$= 8.08 \text{ m/s} \quad \checkmark$$

3 Question 3 / Vraag 3

[4]

A person applies a force $F = (4i - 2j + 7k)$ N to move a crate on a track in the xz -plane that angles 32° above the positive x -axis. Find the work done by the person over the first 3.00 m of motion. HINT: Determine the displacement in unit vector notation.

'n Persoon pas 'n krag van $F = (4i - 2j + 7k)$ N toe op 'n spoor beveg in die xz -vlak wat 'n hoek van 32° maak bo die positiewe x -as. Vind die arbeid verrig deur die persoon tydens die eerste 3.00 m van beweging. WENK: Bepaal die verplasing in eenheidsvektor notasie.



$$\vec{r} = 3.00m \cos 32^\circ \hat{i} + 3.00m \sin 32^\circ \hat{k} \quad \checkmark$$

$$\vec{F} = (4i - 2j + 7k) \text{ N}$$

$$W = \vec{F} \cdot \vec{r} = (4i - 2j + 7k) \text{ N} \cdot (3m \cos 32^\circ \hat{i} + 3m \sin 32^\circ \hat{k}) \quad (2.54) \quad (1.59)$$

$$= 10.17 \text{ J} + 11.12 \text{ J}$$

$$= 21.29 \text{ J}$$

$$= 21.3 \text{ J} \quad \checkmark$$

Question 4 / Vraag 4

[6]

A body of mass 100 g is attached to a hanging spring whose force constant is 10 N/m. The body is lifted until the spring is in its unstretched state. The body is then released. Using the law of conservation of mechanical energy, calculate the speed of the body when it strikes the table 15 cm below the release point.

'n Liggam met massa 100 g is aan 'n veer, wat vertikaal hang, vasgemaak. Die veerkonstante is 10 N/m. Die liggam word opgelig totdat die veer in sy ongerekte posisie is. Die liggam word dan losgelaat. Gebruik die behoud van meganiese energie, en bereken die spoed van die liggam wanneer dit die tafel, 15 cm onder die loslaatpunt, tref.



$$y = 15 \text{ cm} \quad \text{---} \quad y = 0$$

$$y = 0 \quad \text{---} \quad y = -15 \text{ cm}$$

must define where
 $y = 0$ or $U_g = 0$ (sketch/words)

No definition (-1)

①

②

Soln ①

$$U_{gi} = mgh \quad U_{gf} = 0$$

$$U_{si} = 0 \quad U_{sf} = \frac{1}{2} k \Delta y^2$$

$$K_i = 0 \quad K_f = \frac{1}{2} m v^2$$

$$U_{gi} + U_{si} + K_i = U_{gf} + U_{sf} + K_f$$

$$mgh + 0 + 0 = 0 + \frac{1}{2} k \Delta y^2 + \frac{1}{2} m v^2 \quad \checkmark$$

$$v = \sqrt{mgy - \frac{1}{2} k \Delta y^2}$$

$$= \sqrt{(0.10 \text{ kg})(9.8 \text{ m/s}^2)(0.15 \text{ m}) - \frac{1}{2} (10 \text{ N/m})^2 (-0.15 \text{ m})^2}$$

$$= 0.831 \text{ m/s} \quad \checkmark$$

Soln 2

$$U_{gi} = 0 \quad U_{gf} = -mgh$$

$$U_{si} = 0 \quad U_{sf} = \frac{1}{2} k \Delta y^2$$

$$K_i = 0 \quad K_f = \frac{1}{2} m v^2$$

Question 5 / Vraag 5

[6]

Consider a system consisting of a single object with a mass m which is moving through a displacement directed to the right under the action of a net force ΣF , also directed to the right. If the object moves through a displacement $\Delta \vec{r} = (x_f - x_i)\hat{i}$, the net work done on the object by the external net external force $\Sigma \vec{F}$ is

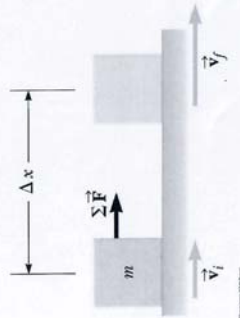
$$W_{\text{ext}} = \int_{x_i}^{x_f} \Sigma F dx.$$

Using this definition for work, show clearly that $W_{\text{ext}} = \Delta K$.

Beskou 'n stelsel met 'n enkele voorwerp met massa m wat beweeg deur 'n verplasing na regs onder die invloed van 'n netto krag ΣF , wat ook na regs gerig is. Indien die voorwerp deur 'n verplasing $\Delta \vec{r} = (x_f - x_i)\hat{i}$ beweeg, word die netto arbeid verrig op die voorwerp deur die netto eksterne krag $\Sigma \vec{F}$ gegee deur

$$W_{\text{ext}} = \int_{x_i}^{x_f} \Sigma F dx.$$

Gebruik hierdie definisie vir arbeid en bewys duidelik dat $W_{\text{ext}} = \Delta K$.



$$W = \int_{x_i}^{x_f} \Sigma F \cdot dx = \int_{x_i}^{x_f} m a \cdot dx \quad [m \text{ is constant}]$$

$$= m \int_{x_i}^{x_f} \frac{dv}{dt} \cdot dx$$

$$= m \int_{x_i}^{x_f} \frac{dv}{dx} \cdot \frac{dx}{dt} \cdot dx \quad [\text{chain rule manipulation}]$$

$$= m \int_{v_i}^{v_f} v \cdot dv$$

$$= m \left[\frac{1}{2} v^2 \right]_{v_i}^{v_f}$$

$$W_{\text{ext}} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 = \Delta K \quad \checkmark$$

[14]

Question 6 / Vraag 6

You throw a 240 g orange straight up in the air from a height of 90 cm above the ground starting from rest. You let go of the orange when it is 140 cm above the ground and it reaches a maximum height of 14 m. Draw a clear sketch showing the relevant positions.

Jy gooi 'n 240 g lemoen in die lug op van 'n hoogte van 90 cm bo die grond waar dit vanuit rus begin. Jy laat los die lemoen wanneer dit 140 cm bo die grond is, en dit bereik 'n maksimum hoogte van 14 m.

Teken 'n skets om al die relevante posities aan te dui.

C $-y_{\max} = 14.0 \text{ m above ground}$ $K_c = 0$, $U_g = mg(\Delta y_2)$

B $-y_i = 140 \text{ cm above ground}$ $K_B = \frac{1}{2}mv^2$, $U_g = 0$

A $-y_i = 90 \text{ cm above ground}$ $K_i = 0$, $U_{gi} = mg(\Delta y_1)$

(a) How much work is done by the force of gravity while your hand is in contact with the orange?
Hoewel arbeid word verrig deur die swaartekrag terwyl jou hand in kontak is met die lemoen? (3)

$$W_g = \vec{F}_g \cdot \vec{d} = -mg \Delta y = -(0.240 \text{ kg})(9.8 \text{ m/s}^2)(0.50 \text{ m}) = -1.176 \text{ J}$$

$$= -1.18 \text{ J}$$

$$= -1.2 \text{ J}$$

(b) How much work is done by you?
Hoewel arbeid word verrig deur jou? (6)

Let $U_g = 0$ at B.

* From B $\rightarrow$ C: Isolated system Orange Earth

$$K_B + U_B = K_C + U_C$$

$$K_B + 0 = 0 + mg \Delta y_2$$

$$= (0.240 \text{ kg})(9.8 \text{ m/s}^2)(14.0 \text{ m} - 1.4 \text{ m})$$

$$K_B = 29.6 \text{ J}$$

* From A $\rightarrow$ B: Non isolated system.

$$W_{\text{ext}} = \Delta K + \Delta U = (K_B - K_A) + (U_B - U_A) = (29.6 \text{ J} - 0) + (0 - (0.240 \text{ kg})(9.8 \text{ m/s}^2)(-0.50 \text{ m})) = 29.6 \text{ J} + 1.176 \text{ J} = 30.8 \text{ J}$$

$$W_{\text{ext}} = 29.6 \text{ J} + 1.2 \text{ J}$$

$$= 30.8 \text{ J}$$

(c) Determine the power delivered by the force of gravity from the time the orange leaves your hand till it reaches its maximum height?

Bepaal die druwing gelewer deur die swaartekrag gedurende die tyd wat die lemoen jou hand verlaat totdat dit die maksimum hoogte bereik? (5)

$$K_B = 29.6 \text{ J} = \frac{1}{2}mv^2$$

$$\therefore v_B = \sqrt{2(29.6 \text{ J}) / 0.240 \text{ kg}} = 15.7 \text{ m/s}$$

$$v_c = 0 \text{ m/s} \quad a_y = -9.8 \text{ m/s}^2$$

$$\Delta t = ?$$

$$v_c = v_B + a_y t$$

$$t = (v_c - v_B) / a_y = (0 - 15.7 \text{ m/s}) / (-9.8 \text{ m/s}^2) = 1.6 \text{ s}$$

$$W_g = \Delta K_{B \rightarrow C} = -29.6 \text{ J}$$

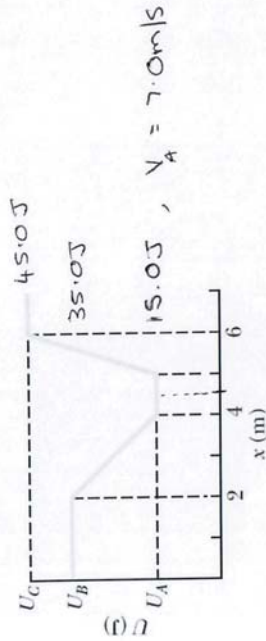
$$P = W_g / \Delta t = -29.6 \text{ J} / 1.6 \text{ s}$$

$$= -18.5 \text{ W}$$

7 Question 7 / Vraag 7

The figure shows a plot of potential energy U versus position x of a 0.90 kg particle that can travel only along an x axis. (Nonconservative forces are not involved.) Three values are $U_A = 15.0$ J, $U_B = 35.0$ J, and $U_C = 45.0$ J. The particle is released at $x = 4.5$ m with an initial speed of 7.0 m/s, headed in the negative x direction.

Die figuur dui die grafiek van potensiële energie U teenoor posisie x van 'n 0.90 kg deeltjie wat slegs langs die x -as kan beweeg. (Gien nie-konservatiewe kragte is betrokke nie.) Die drie waardes is $U_A = 15.0$ J, $U_B = 35.0$ J en $U_C = 45.0$ J. Die deeltjie word by $x = 4.5$ m losgelaat met 'n aanvanklike speed van 7.0 m/s, gerig in die negatiewe x -rigting.



(a) If the particle can reach $x = 1.0$ m, what is its speed there, and if it cannot, what is its turning point?

Indien die deeltjie $x = 1.0$ m kan bereik, wat is die speed hier? Indien die deeltjie nie $x = 1.0$ m kan bereik nie, waar is die draaipunt?

$$U_A + K_A = U_B + K_B$$

$$15.0 \text{ J} + \frac{1}{2}(0.90 \text{ kg})(7.0 \text{ m/s})^2 = 35.0 \text{ J} + \frac{1}{2}(0.90 \text{ kg})v_B^2$$

$$\frac{1}{2}mv_B^2 = 15.0 \text{ J} + 22.1 \text{ J} - 35.0 \text{ J} = 2.1 \text{ J}$$

$$v_B = \sqrt{2(2.1 \text{ J}) / 0.90 \text{ kg}}$$

$$v_B = 2.2 \text{ m/s}$$

(b) What are the magnitude and direction of the force on the particle as it begins to move to the left of $x = 4.0$ m? Wat is die grootte en rigting van die krag op die deeltjie soos dit by $x = 4.0$ m na links begin beweeg?

$$|F| = \frac{dU}{dx} = (35 - 15) \text{ J} / (4.5 - 4.0) \text{ m}$$

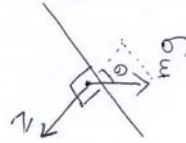
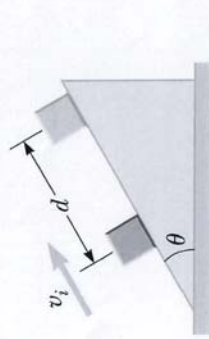
$$= -5.7 \text{ N}$$

$$|F| = 5.7 \text{ N} \text{ in the } +x \text{ direction}$$

8 Question 8 / Vraag 8

A 5.00 kg block is set into motion up an inclined plane with an initial speed of $v_i = 8.00$ m/s. The block comes to rest after travelling $d = 3.00$ m along the plane which is inclined at an angle of $\theta = 30.0^\circ$ above the horizontal. For this motion determine the following:

'n 5.00 kg blok beweeg by 'n skuinshlak op met 'n aanvanklike speed van $v_i = 8.00$ m/s. Die blok kom tot rus nadat dit 'n afstand $d = 3.00$ m langs die skuinshlak (wat teen 'n hoek van $\theta = 30^\circ$ bo die horisontaal gerig is) beweeg het. Vir die beweging bepaal die volgende:



- (a) the change in the block's kinetic energy.
die verandering in die blok se kinetiese energie.

(2)

$$\Delta K = K_f - K_i = 0 - \frac{1}{2}mv_i^2 \quad \checkmark$$

$$= -\frac{1}{2}(5.00 \text{ kg})(8.00 \text{ m/s})^2 \quad \checkmark$$

$$= -160 \text{ J} \quad \checkmark$$

- (b) the change in the potential energy of the block - Earth system.
die verandering in die potensiele energie van die blok - Aarde stelsel.

(2)

$$\sin \theta = \frac{\Delta y}{d}$$

$$\therefore \Delta y = d \sin \theta$$

$$= 3.00 \text{ m} \sin 30^\circ = 1.5 \text{ m}$$

$$\Delta U = mg \Delta y = (5 \text{ kg})(9.8 \text{ m/s}^2)(1.5 \text{ m})$$

$$= 73.5 \text{ J}$$

- (c) the coefficient of kinetic friction (assuming the friction force is constant).
die kinetiese wrywings-koeffisiënt (neem aan dat die wrywingskrag konstant is).

(4)

$$\Delta K + \Delta U = W_{\text{other forces}} - f_k d \quad \checkmark$$

$$= 0 - f_k d = -\mu_k N d \quad \checkmark$$

$$\mu_k = -(\Delta K + \Delta U) / N d \quad \checkmark$$

$$= -(-160 \text{ J} + 73.5 \text{ J}) / (mg \cos 30^\circ)(3.00 \text{ m})$$

$$= 0.68 \quad \checkmark$$

The end! / Die einde!